

DOCUMENT DESIGN & GRAPHICS

How words look matters as much as what they say.

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Foundations of Design

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Big Idea

Imagine the challenge of searching for an item in a disorganized room with poor lighting. It's comparable to searching for a needle in a haystack. Similarly, a disordered document can lead to confusion among readers who are attempting to find specific information.

- Document design is about organizing and presenting information on a page.
- It's about how layout, text, and visuals work together.
- Design impacts reader comprehension and actions.

Chapter Goals

By the end of this chapter, you will be able to:

- Design documents that are tailored to a specific audience and purpose
- Select the graphic that best fits the information you're presenting
- Format graphics so they communicate clearly and effectively
- Construct your own tables, graphs, charts, and diagrams

Why Design Matters

- In technical writing, presentation carries as much weight as content
- Readers judge a document's clarity within seconds of seeing it, before reading a single word
- Well-designed documents reduce the reader's effort, allowing them to focus on the content rather than deciphering it

Below is an example of instructions of planting tea rose with both bad and good design

Tea Rose

Find an area that receives full sun. Use rich, well-drained soil that contains organic matter. Dig a hole slightly wider than the root-ball. Plant the bush so that the top of the root-ball is exposed. Do not cover the root-ball. Water the area thoroughly and fertilize twice a year: in spring and in late summer.

Tea Rose

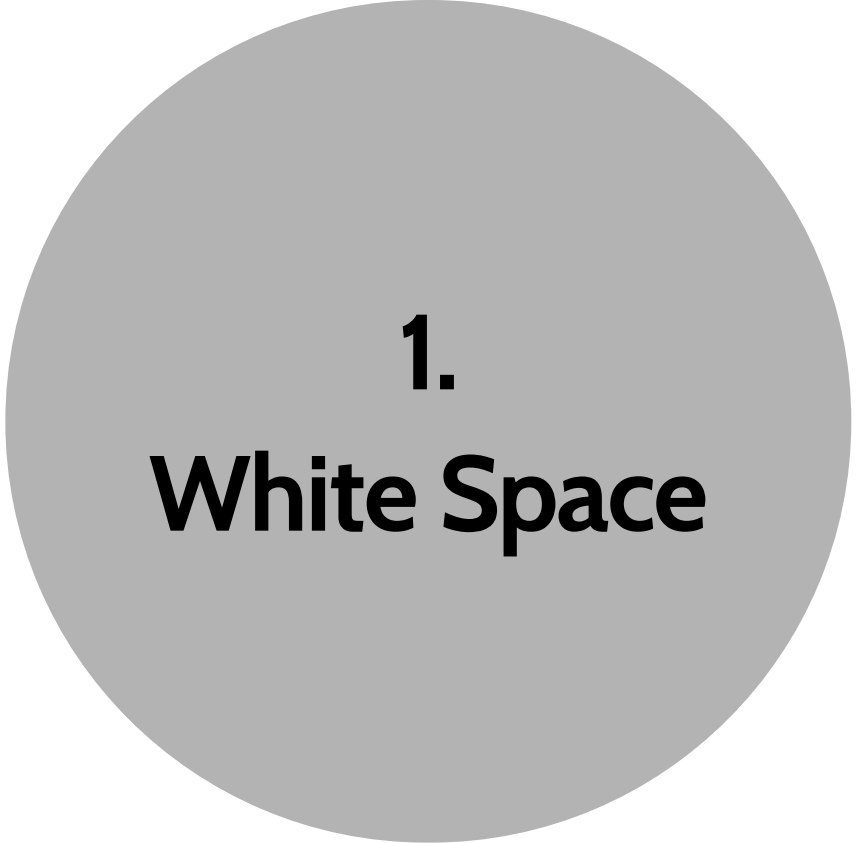
1. Find an area that receives full sun.
2. Use rich, well-drained soil that contains organic matter.
3. Dig a hole slightly wider than the root-ball.
4. Plant the bush so that the top of the root ball is exposed. Do not cover the root-ball.
5. Water the area thoroughly.
6. Fertilize twice a year: in spring and in late summer.

figure 1: Examples of Poorly Designed and Well-Designed Instructions

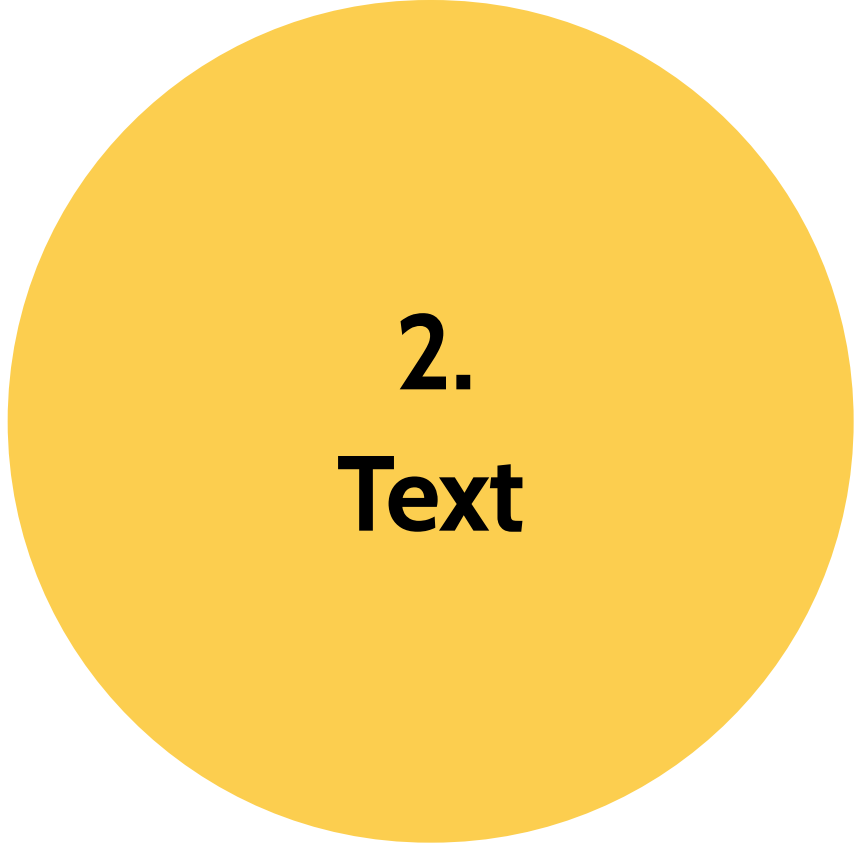
Design Elements

Design elements play a crucial role in the composition of a page. They encompass key components such as white space, text, headings, and graphics. These elements are meticulously organized by a writer to not only determine the visual appeal but also enhance readability.

Four design elements are:



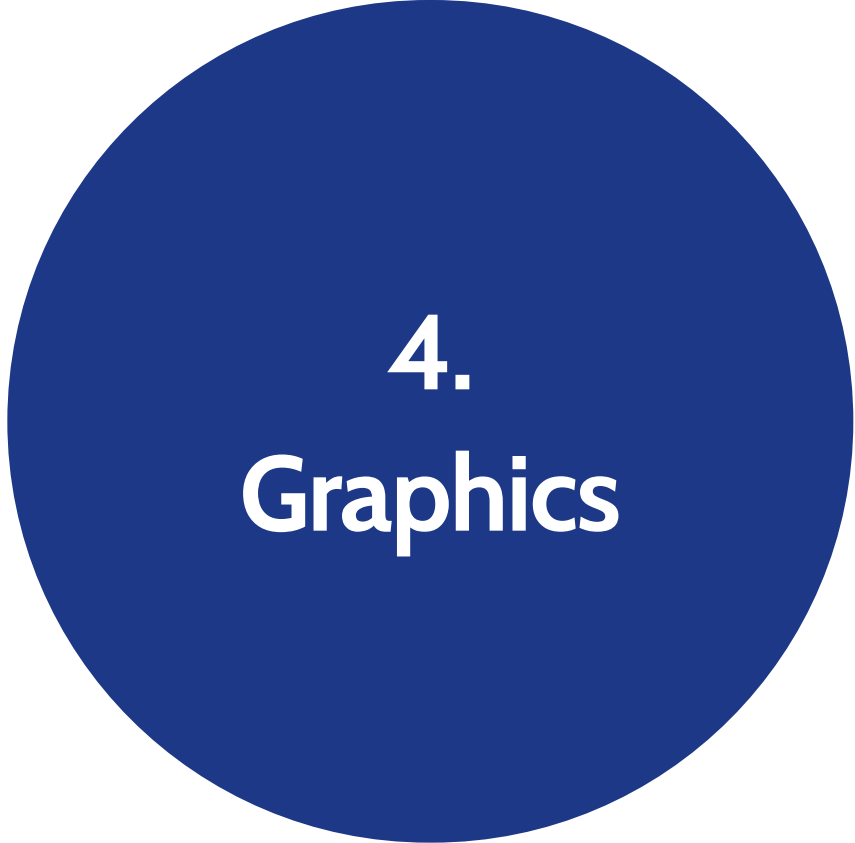
1.
White Space



2.
Text



3.
Headings



4.
Graphics

1. White Space

- Blank space with no text or graphics
- Rests the eyes and separates chunks of information
- Makes a document look inviting rather than crammed
- Used in margins, between paragraphs, around lists and graphics

2. Text

- Left-justified, ragged-right text is read fastest
- Serif fonts (with tails) are easy to tell apart — good for body text
- Sans serif fonts are clean and bold — good for titles and headings
- Body text: 10–12 pt; avoid ALL CAPS except for emphasis or titles

3. Heading

- Short titles that show the reader your organization at a glance
- Work like a formal outline — size and placement show hierarchy
- Most documents use two levels; some use more

4. Graphics

- Visual representations of information: tables, graphs, charts, diagrams, and more
- Used whenever information communicates better visually than in words
- Can stand alone (traffic signs) or work with text

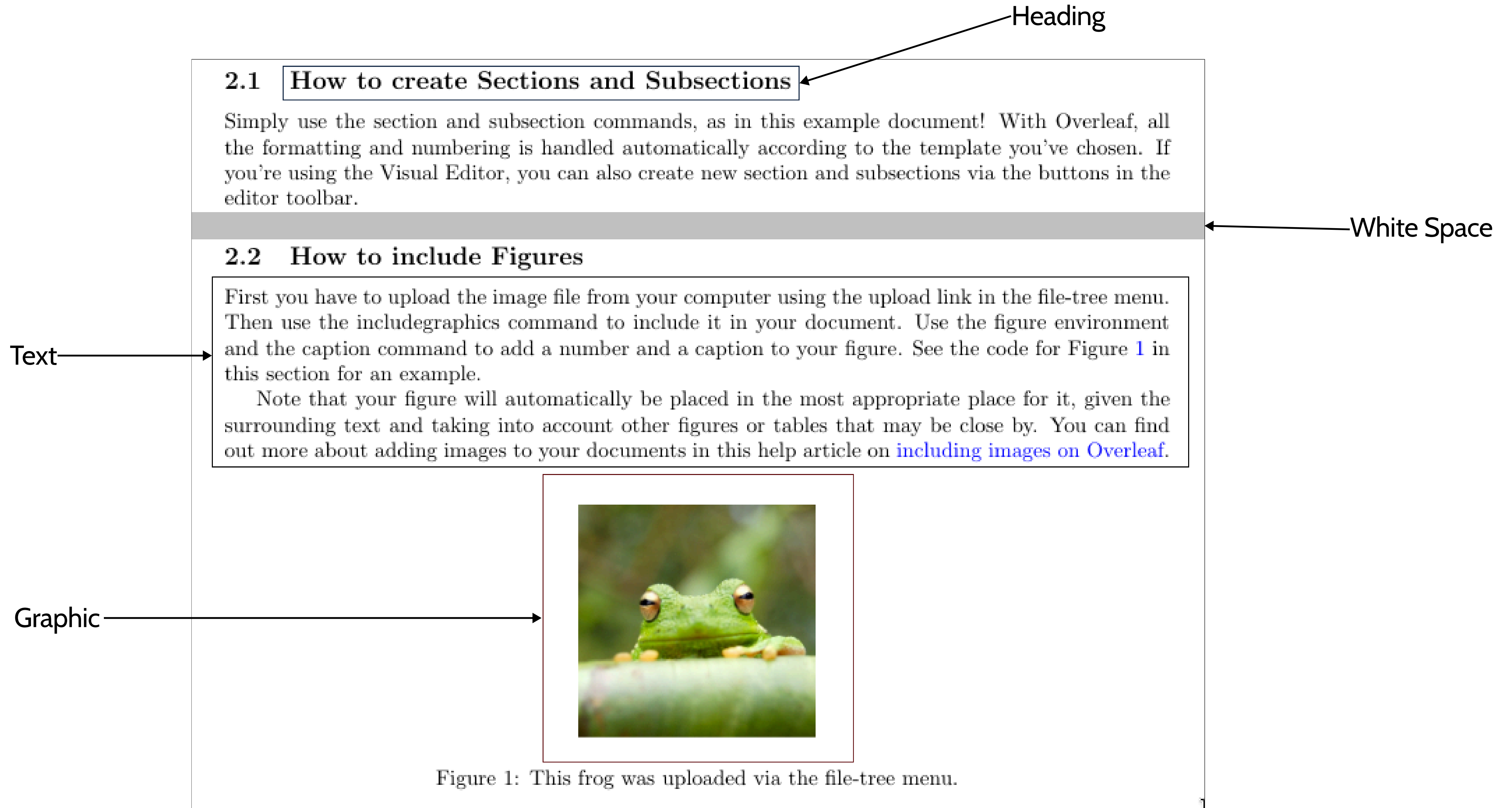


figure2: Example of design elements in documents

Choosing & Designing Graphics

 Sudip Khadka

What is Graphics?

Graphics play a crucial role in conveying information effectively. They encompass a variety of visual elements such as tables, graphs, charts, and diagrams, which are utilized to complement or substitute textual content.

- Graphics are used when information is easier to grasp visually than in words
- They can stand alone, like a traffic sign, or work together with surrounding text
- The right graphic doesn't just illustrate information — it makes it easier to understand

Choosing a Medium

- Medium refers to how a document is delivered, including print, digital screens, or a mix.
- The choice between paper and electronic formats depends on your audience and purpose
- A printed mailer might reach older or local readers; a digital post reaches a wider audience faster
- The medium you choose shapes every design decision that follows, from layout to graphic style

Who reads graphics?

Understanding the audience is crucial before choosing a graphic element. Data interpretation and presentation can differ based on the target demographic.

Answering following questions can help identify who reads the graphics



How much does my reader already know?



How interested are they in the subject?



Do they need visuals or will visuals confuse them?



What's their reading level, and how will they access graphics?

Purpose: Matching Graphic to Goal

The purpose of a graphic is crucial as it determines whether it is used to compare numbers, illustrate a trend, or clarify a process.

To help you select an appropriate graphic, you can use following table as a guide

Purpose	Graphic
Small amounts of data	an informal table
Numbers that need close comparison	a formal table
A trend over time	a line graph
Parts of a whole	a pie graph
A step-by-step process	a flowchart
An organization's structure	an organization chart

4 Rules For Any Graphic

Graphics play a crucial role in helping people grasp information more easily. Let's dive into the four rules designed to ensure they do just that:

Rule 1: Neat and Simple

- One point per graphic — avoid cluttered ideas
- Leave white space, ensure readability
- Align decimals in columns

Rule 2: Integrate with text

- Refer to each graphic by number before displaying: 'Table 1 shows...'
- Use 'figure' for non-formal content and 'table' for formal tables.
- Use small graphics near references; large ones may require an appendix.
- Highlight the importance, not every detail

Rule 3: Give Credit

- Title every graphic (except informal tables)
- Use specific titles: 'Salary Distribution 2015–2017,' not just 'Salaries'
- Add a Source line with citation if data isn't your own
- Note if reprinted with permission or adapted

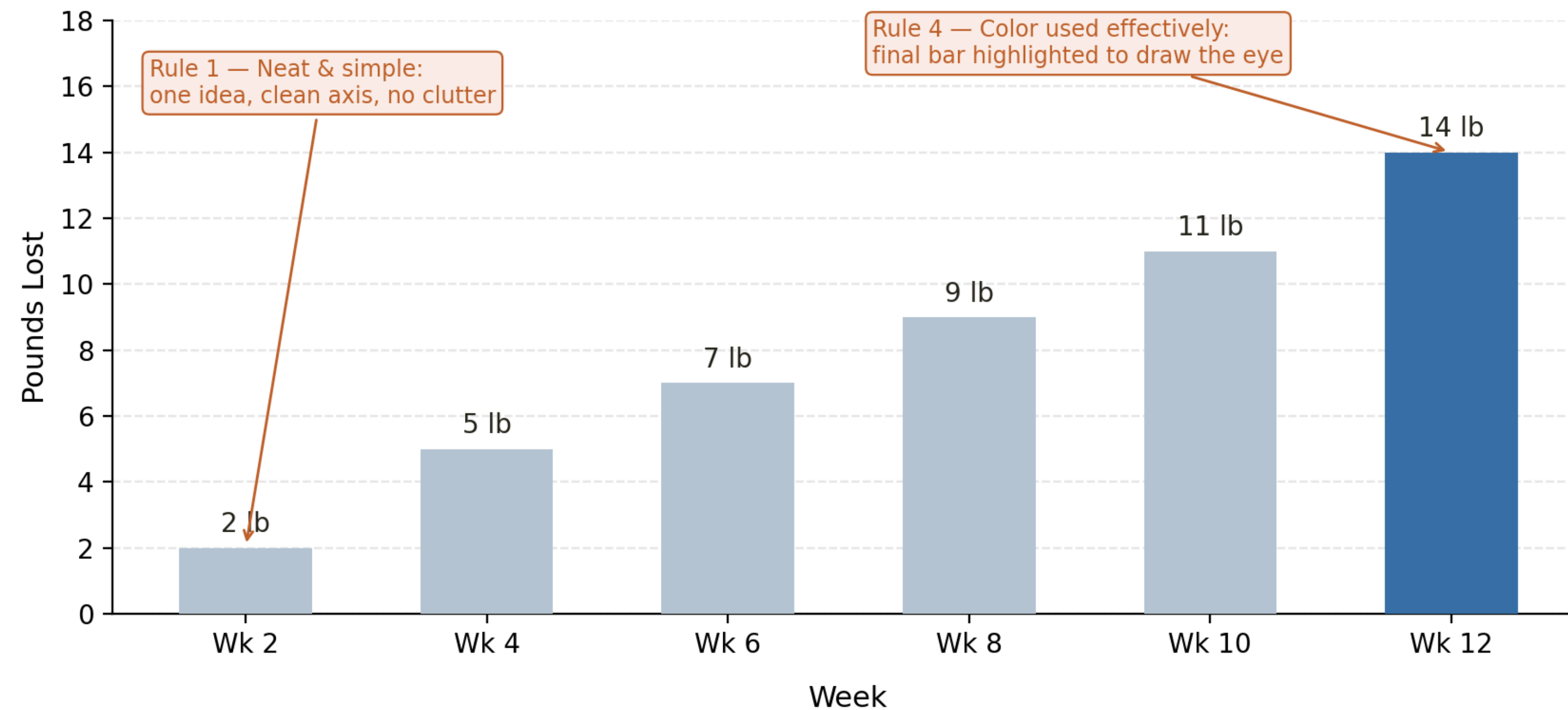
Rule 4: Use color effectively

- Shows a document's organization; emphasizes key points
- Reinforces meaning (red = danger) and improves visibility
- Max 3–4 colors; stay consistent throughout
- Colors carry different cultural meanings — check your audience

As Figure 3 shows, participants lost weight steadily, with the sharpest gain at the end.

Rule 2 — Integrated with text: referenced as "Figure 3" above

Figure 3: Weight Lost per Two-Week Interval



Source: Sample fitness program data, internal tracking, 2026.

Rule 3 — Credit given: titled clearly, source cited below

figure3: Example showing 4 rules applied to the graphic

Tables & Graphs



Anush Shrestha

Tables

Tables are used to present words or numbers organized into categories of columns and rows.
Types of Table are discussed below:

Informal Table

- Rows/columns with no lines, flows with text
- Best for simple info, younger/less technical audiences
(Example: daily calorie needs for teen boys vs. girls)

Verbal Table

- Like a formal table, but uses words instead of numbers
- Good for showing relationships simply
(Example: diseases linked to poor diet)

Formal Table

- Rows/columns with rules (lines), used for detailed numeric data
- Best for knowledgeable audiences who can interpret data independently
(Example: recommended heart rates by age)

Tables: Examples

Informal Table Example

The Dietary Guidelines for Americans recommends the following calorie needs for moderately active boys and girls aged 13:

Daily Calories for Boys 2200

Daily Calories for Girls 2000

figure4: Model of Informal Table

Formal Table Example

Age	Maximum Heart Rate	Target Zone(50-80%)
20	200	100-170 bpm
40	180	90-153 bpm
60	160	80-136 bpm

Table 1: Recommended Heart Rates by Age

Source: American Heart Association guidelines.

Verbal Table Example

Disease	Diet Factor
Heart Disease	High Saturated fat intake
High Blood Pressure	Excess Sodium Consumption
Type 2 Diabetes	High Sugar and refined carbohydrate intake

Table 2: Diseases Linked to Poor Eating Habits

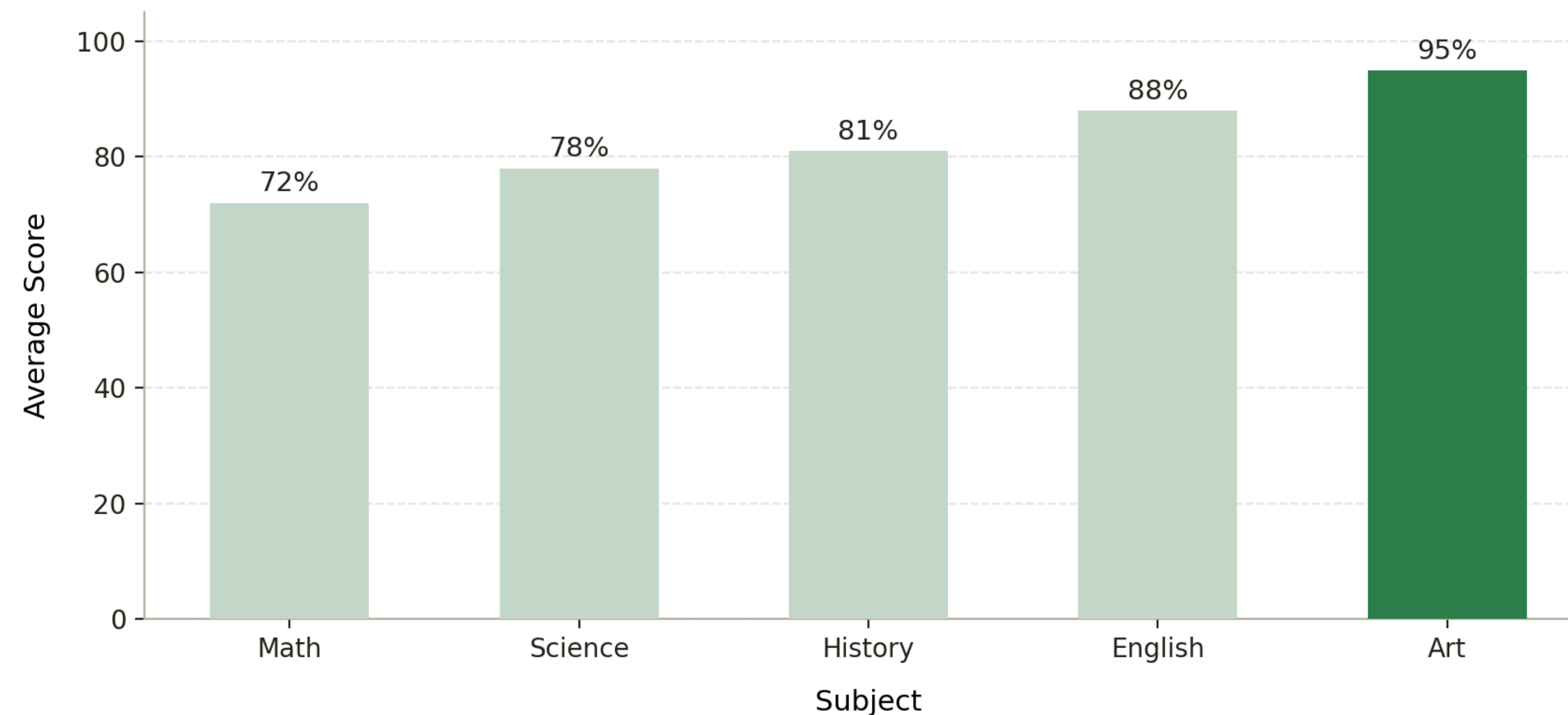
Bar Graph

The bar graph is a graph that uses a horizontal axis and a vertical axis to compare numerical data presented in rectangular columns or bars. The bars can run vertically or horizontally.

- Horizontal = more room for long labels
- Vertical axis = dependent variable; horizontal axis = independent variable (often time)

As Figure 5 shows, students performed strongest in Art, with steady scores across other subjects.

Figure 5: Average Test Scores by Subject



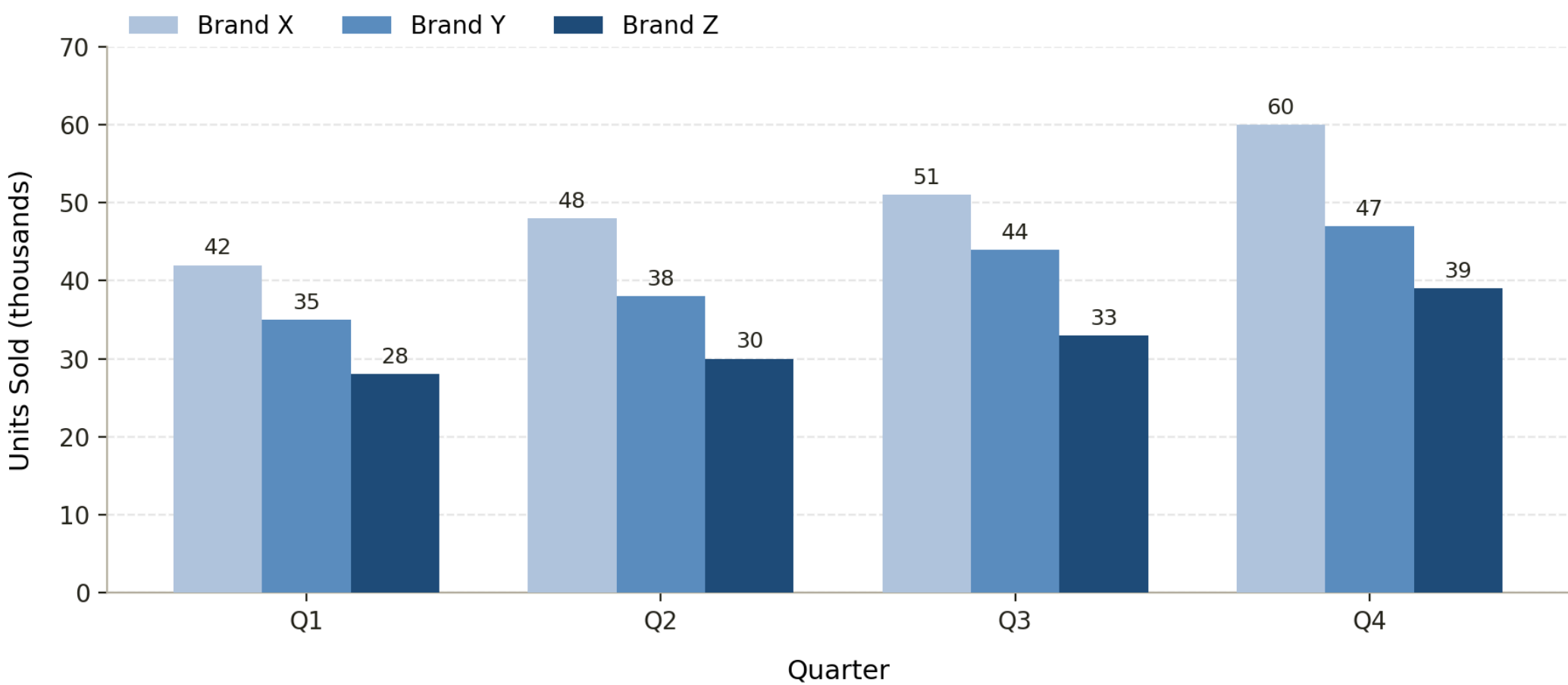
Source: Sample classroom assessment data, spring term, 2026.

Multiple Bar Graph

- A multiple bar graph compares sets of data by placing bars side by side within each group.
- Each bar within a cluster represents a different data series, distinguished by color
- A legend or key is essential — without it, readers can't tell which bar represents which data set

As Figure 6 shows, Brand X led sales every quarter, while all three brands trended upward through the year.

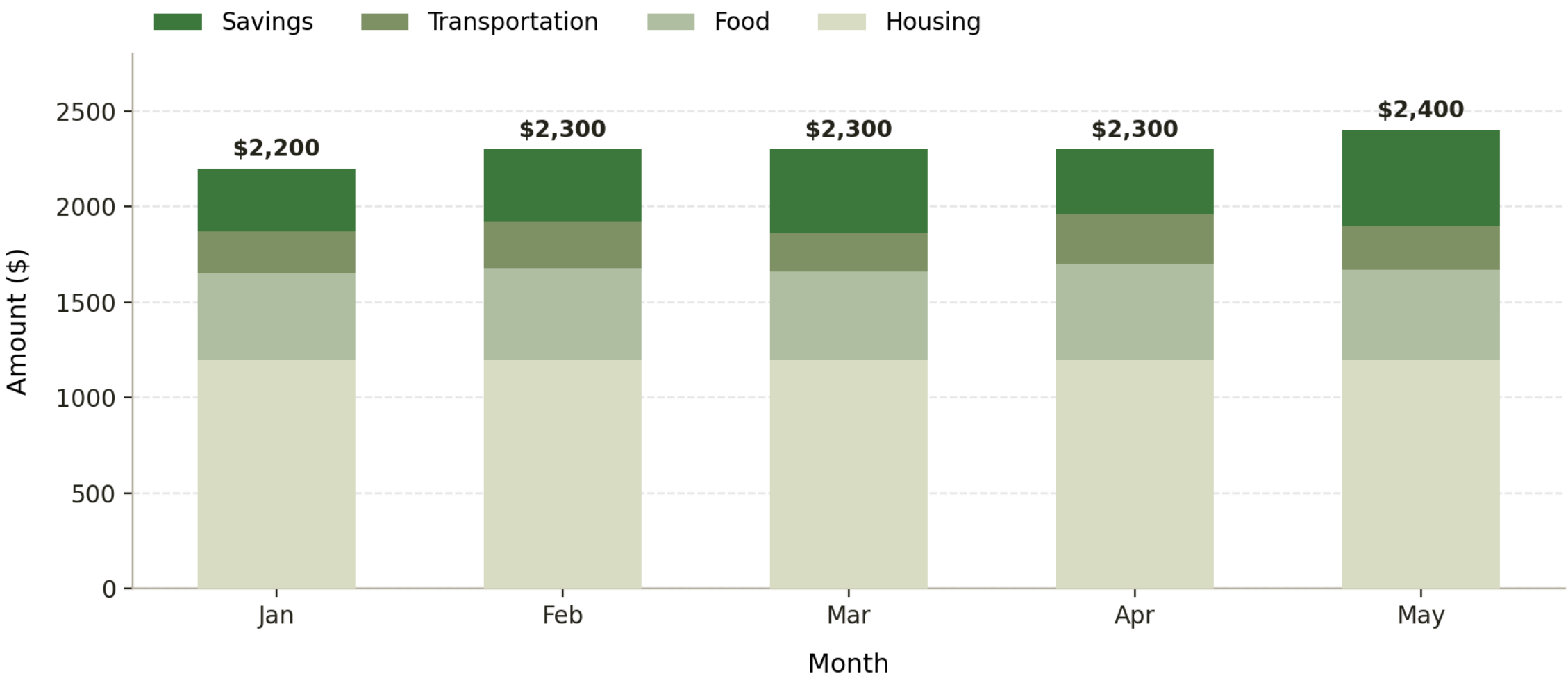
Figure 6: Quarterly Unit Sales by Product Line



Source: Sample regional sales data, fiscal year 2026.

As Figure 7 shows, housing remains the largest fixed cost, while savings grew steadily across the first five months.

Figure 7: Monthly Household Budget Breakdown



Source: Sample household budget data, Jan-May 2026.

Divided Column Graph

- It stacks data segments in bars to show how parts make up a whole.
- Each column represents one total; the segments within it show what that total is made up of
- Useful for displaying overall trends across categories and individual category compositions simultaneously.
- Stack segments consistently in each column for easy comparison (e.g., "Fat" at the bottom, "Muscle" in the middle, "Bone" on top)

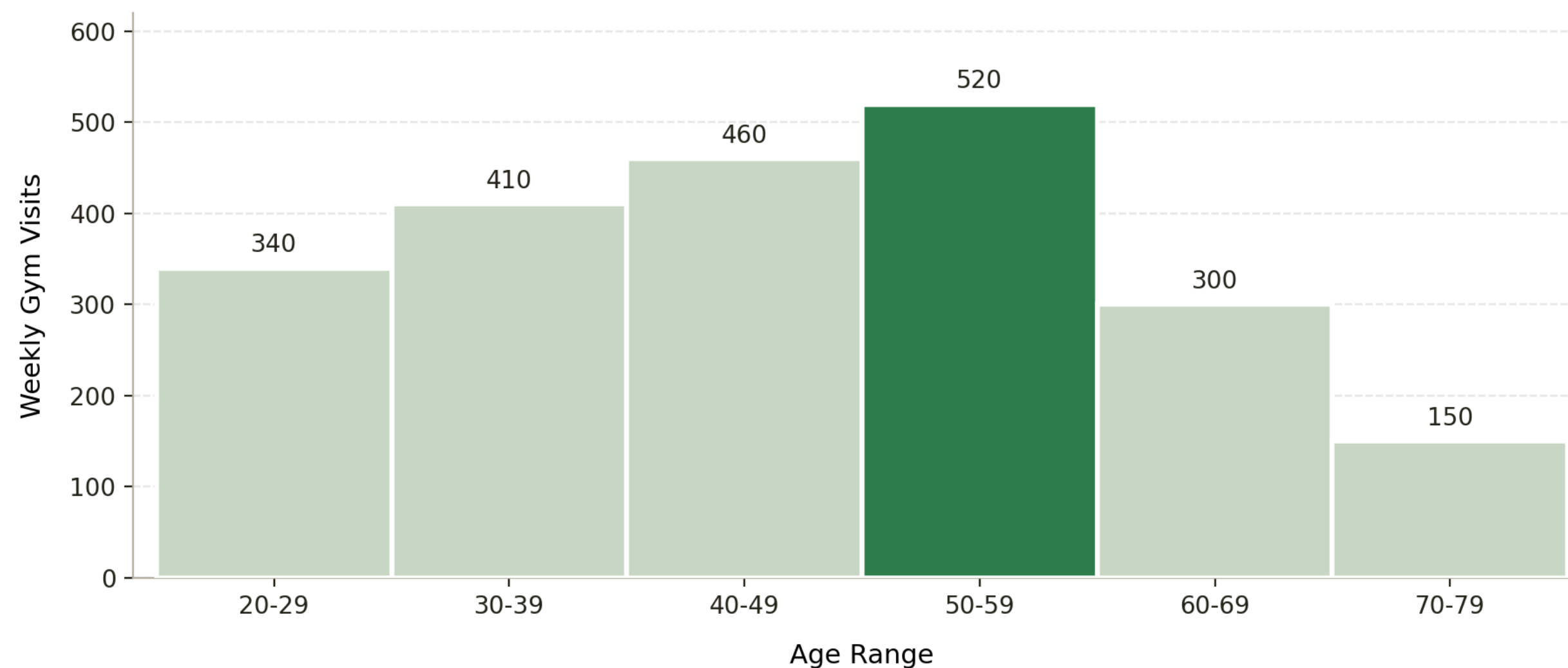
Histogram

A histogram groups continuous numerical data into ranges (called "bins") rather than showing separate, discrete categories.

- Bars in a histogram sit flush with no gaps, indicating continuous ranges, distinguishing it from a bar graph.
- Ideal for displaying frequency or distribution.
- Bin width matters: aim for 5–8 bins in most cases.
- Histogram shape can quickly show patterns like bell curve or skew, hidden in number tables.

As Figure 8 shows, gym use peaks among members aged 50-59, then declines steadily in older age ranges.

Figure 8: Weekly Gym Visits by Age Range



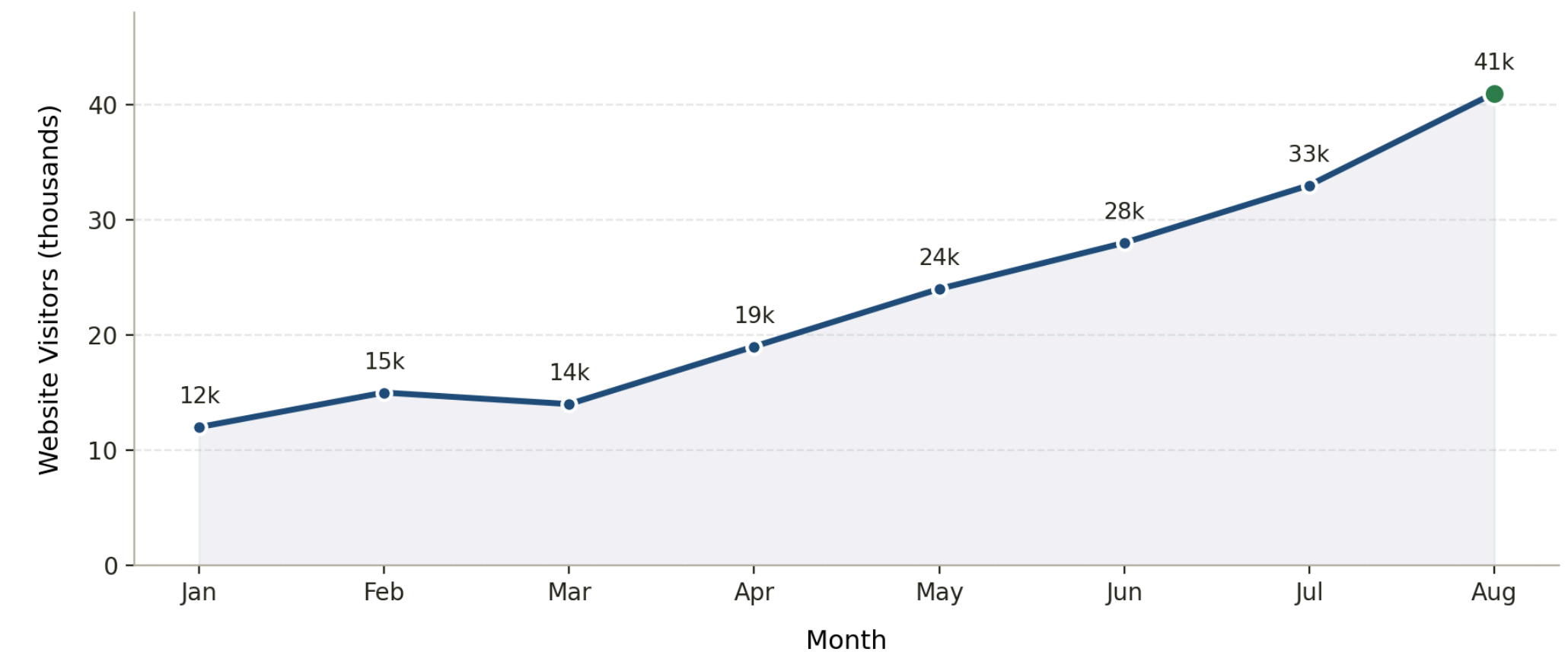
Source: Sample fitness center membership data, 2026.

Line Graphs

- Shows a trend over time rather than individual numbers
- Multiple line graph compares several trends at once
- Avoid clutter for readability
- The line shows the direction and rate of change at a glance, not just individual values.
- Markers at each data point help readers locate exact values without cluttering the line itself

As Figure 9 shows, website traffic grew steadily through the year, more than tripling from January to August.

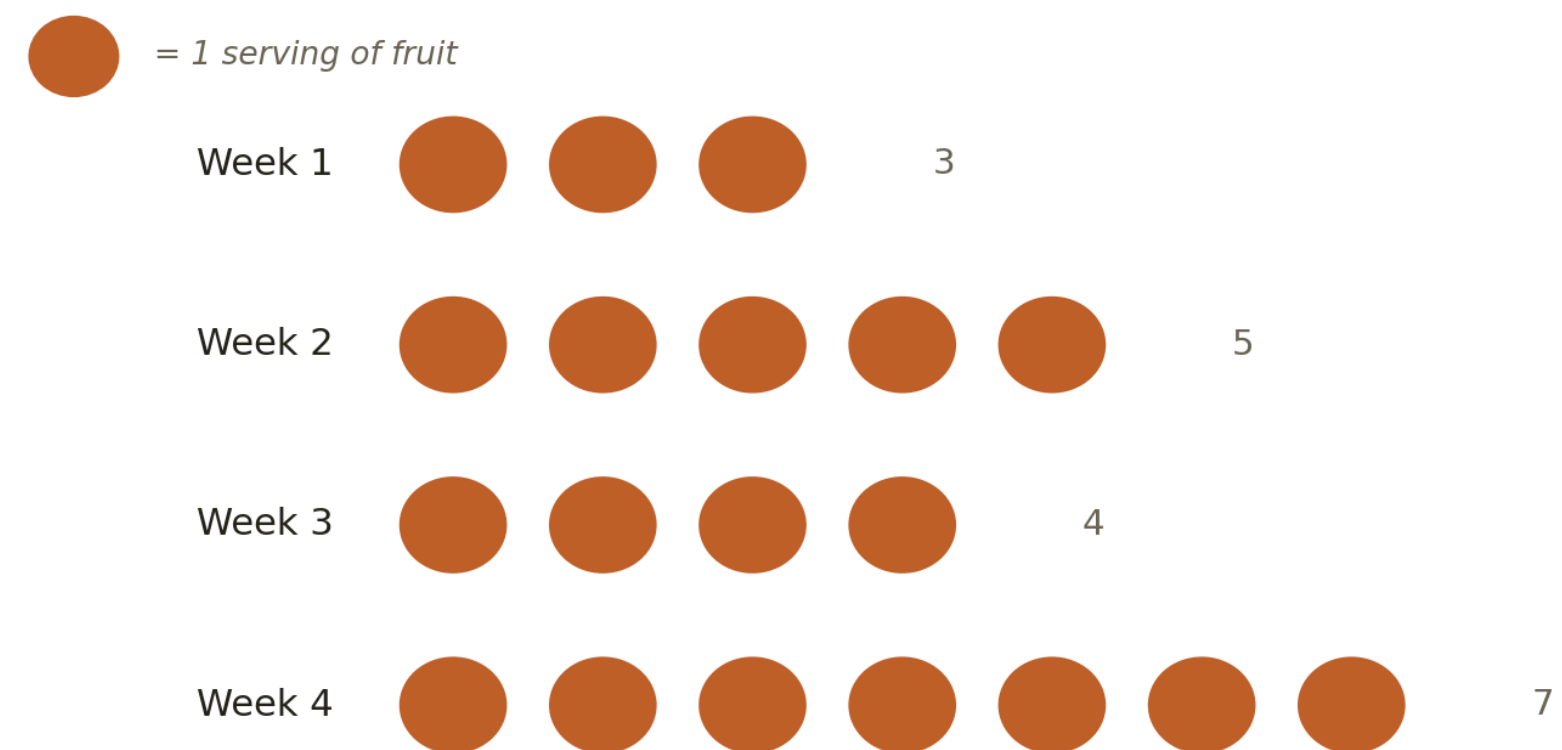
Figure 9: Monthly Website Traffic Growth



Source: Sample web analytics data, Jan-Aug 2026.

As Figure 10 shows, fruit intake rose and fell week to week, peaking at 7 servings in Week 4.

Figure 10: Weekly Fruit Servings, Household Average



Source: Sample household nutrition log, 2026.

Pictograph

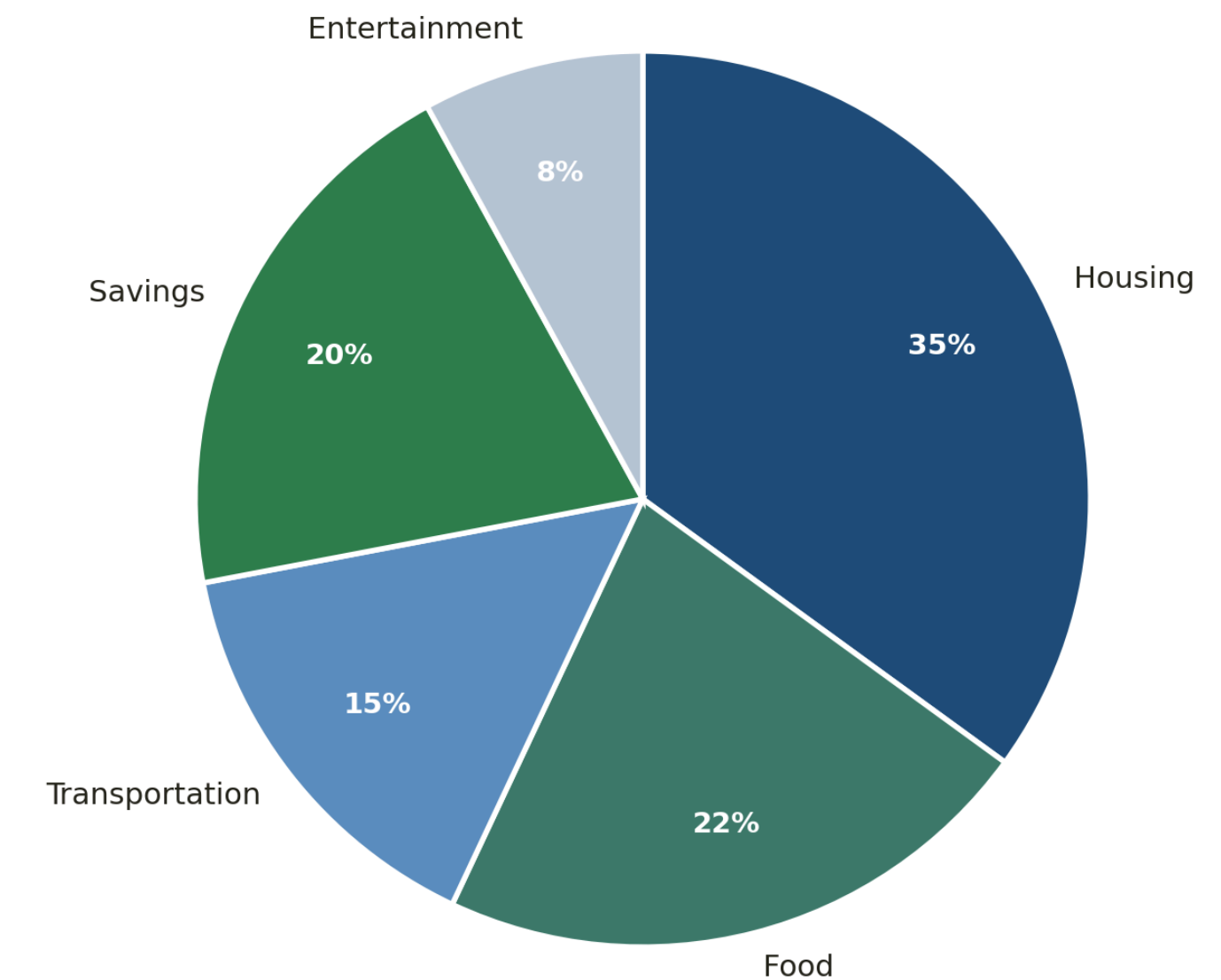
- A pictograph is essentially a bar graph that uses repeated icons instead of bars to represent quantity
- Key: "each icon = X units" is essential for meaningful graphics.
- Use partial icons (e.g., half an icon) to represent partial values when precision matters
- Simple, recognizable icons are key
- Great for rough comparisons, not precise data. Pair with numeric labels for accuracy.

Pie Graphs

- A pie graph illustrates the relationship of parts to a whole, with the full circle always equaling 100%.
- Limit to 7 slices max to avoid unreadable smaller slices.
- Arrange slices largest to smallest, clockwise from 12 o'clock, for natural reading order.
- Label each slice with its percentage directly, avoiding the need for color matching to a separate legend.

As Figure 11 shows, housing accounts for over a third of the monthly budget, more than any other category.

Figure 11: Monthly Budget Allocation by Category



Source: Sample household budget data, 2026.

Infographics

- An infographic combines text and graphics into a compact visual to teach concepts quickly to a broad audience.
- Infographics allow for more color and bold design, aiming for visual appeal and quick comprehension, rather than restraint.
- Break content into clear, bite-sized sections like numbered steps, categories, or stats instead of one dense block.

FOUR TYPES OF GRAPHICS AT A GLANCE

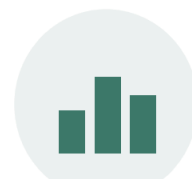
Choosing the right family starts with knowing what each one does best



TABLES

Show Exact Numbers

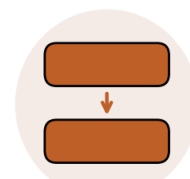
Best for precise data readers need to compare row by row.



GRAPHS

Show Trends & Comparisons

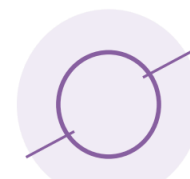
Bar, line, and pie graphs reveal patterns numbers alone can't show.



CHARTS

Show Process & Structure

Flowcharts and org charts map out steps, decisions, and hierarchy.



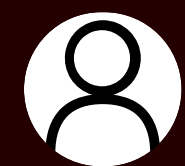
DIAGRAMS

Show How Something Looks

Labeled drawings and icons explain form and function at a glance.

Source: Adapted from Document Design & Graphics, Ch. 6, 2026.

Charts & Diagrams

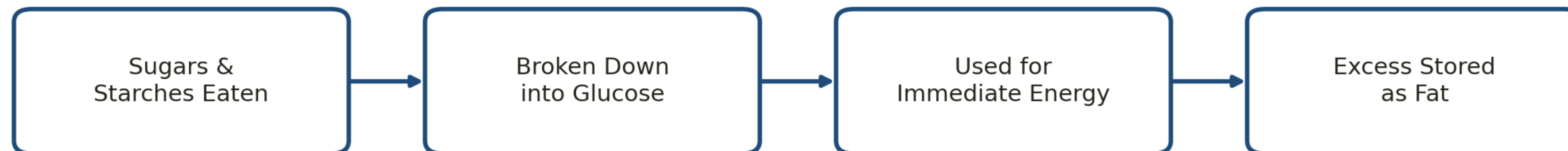


Upendra Raj Panta

Flowcharts

- A flowchart visually represents a process from start to finish using connected boxes and arrows.
- Ideal for readers new to the topic, as it simplifies a complex process into easy-to-follow steps instead of a dense block of text.

Figure 12: How the Body Converts Sugar to Fat

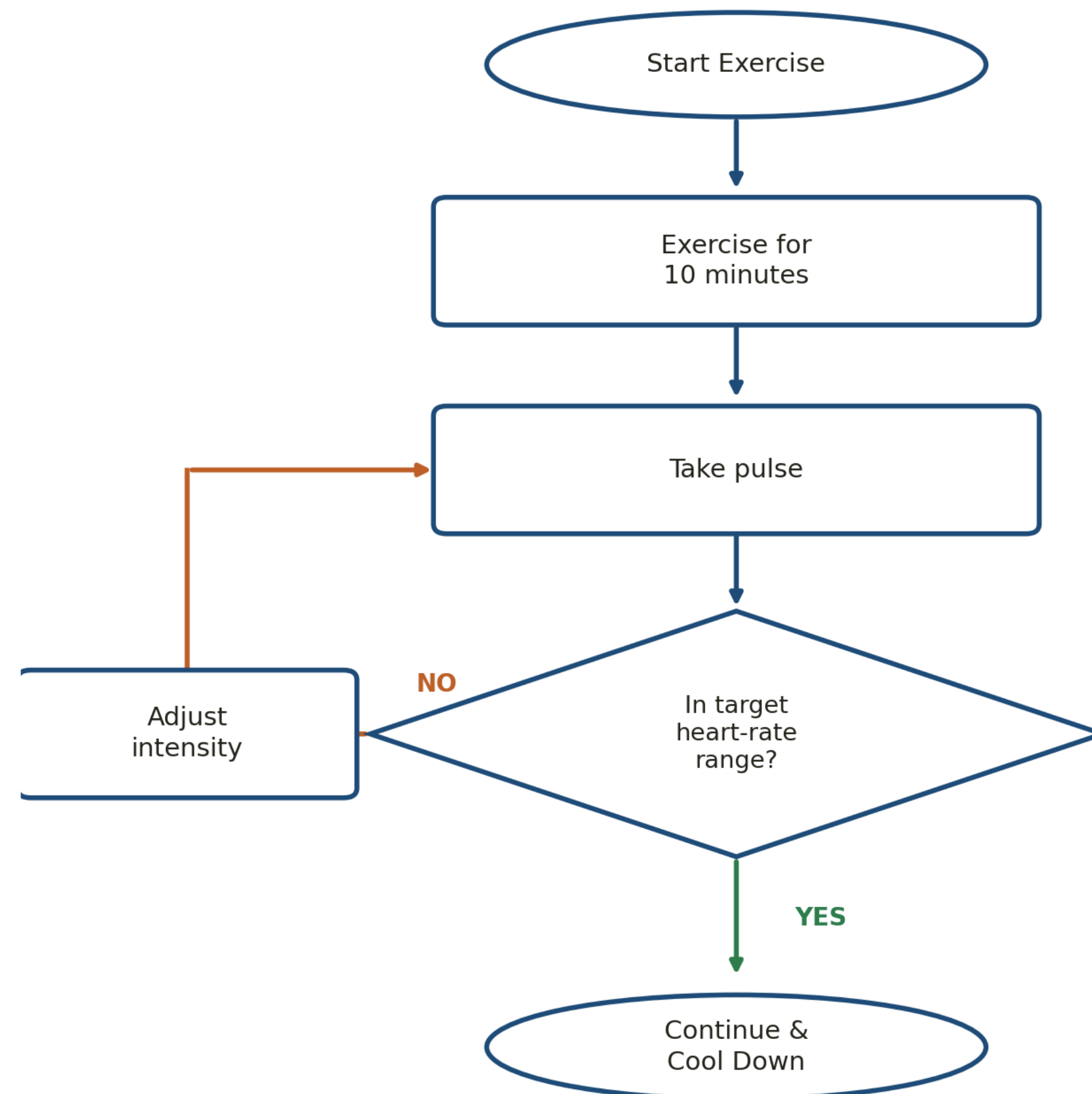


Source: Adapted from general nutrition science references, 2026.

Decision Flowcharts

- A specialized flowchart that uses standard symbols so any reader can interpret it without a key.
- An oval marks start/end, a rectangle marks an action, and a diamond marks a decision point with branching outcomes (usually Yes/No)
- Focus on key steps only; detailed decision flowcharts become unreadable quickly.

Figure 13: Decision Flowchart for Target Heart-Rate Exercise

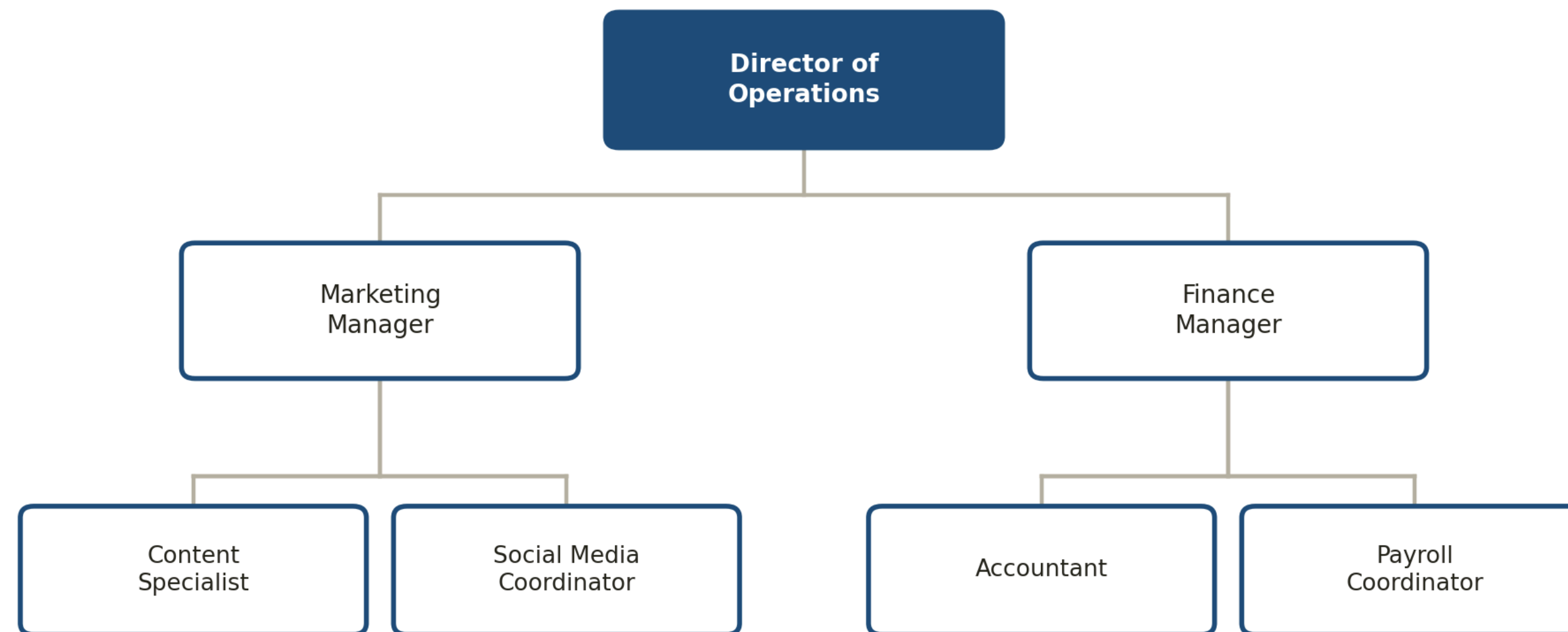


Source: Adapted from general fitness guidelines, 2026.

Organizational Chart

- Boxes and lines display hierarchy within an organization.
- The highest position sits at the top, with each tier below.
- Lines represents reporting relationship, not job duties.

Figure 14: Organizational Chart, Operations Department

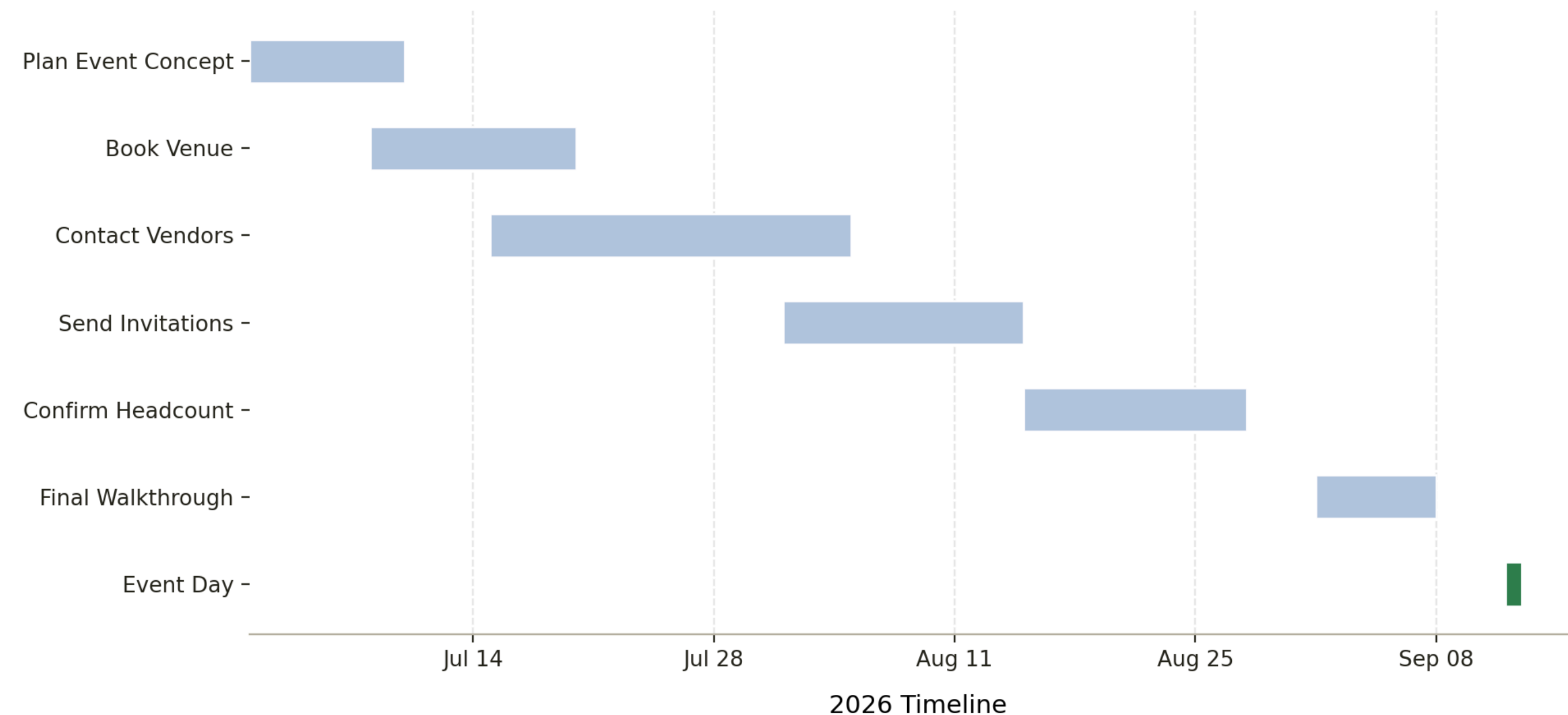


Source: Sample company organizational structure, 2026.

Gantt Chart

- A specialized bar graph used to schedule the tasks of a project over time.
- Each task gets a horizontal bar spanning its start and end dates.
- Shows overlaps, dependencies, and the timeline clearly, which a text schedule can't do as effectively.

Figure 15: Event Planning Schedule



Source: Sample event planning timeline, 2026.

Diagram & Icons

Diagrams

- A diagram is a simplified line drawing that highlights essential parts.
- Diagrams use callouts: leader lines connect labels to parts of the drawing, allowing precise labeling without clutter.

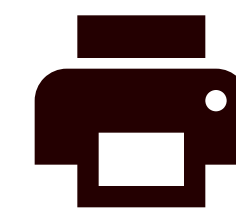
Icons

- An icon is a simple symbol representing an idea, like a wheelchair for accessibility or a printer for printing.
- Icons don't require captions; they are designed for instant understanding without any text.

Figure 16: Anatomy of a Well-Designed Page



Example of Icon Set



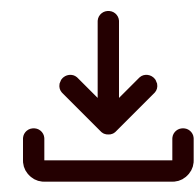
Print



Accessibility



Wifi



Download

Photographs

- A photo's value lies in capturing precise appearance or emotion, which a chart or icon cannot mimic.
- Its design job isn't to invent data, but to be placed and credited correctly.
- Photo should be clear, focused on one idea, and avoid busy backgrounds.
- Crop tightly for focus on the photo's point
- Always include a caption describing what's shown
- Always get permission before publishing a photo of an identifiable person. Note this explicitly if it's not your own.



Source: [unsplash.com](https://unsplash.com/photos/Batuhan-Doğan)-[[Batuhan Doğan](#)], 2026

KEY TAKEAWAYS

- Design plays a critical role beyond decoration; it is the key factor that influences the effectiveness of delivering your message to the audience.
- It is essential to ensure that every decision made is carefully aligned with the specific needs and objectives of the target audience.
- When presenting information, it is crucial to maintain simplicity, provide proper attribution to your sources, and ensure the accuracy of your data without any manipulation.

THANK

YOU!!!!